

Computer Vision – Lecture 3

Image Processing II

14.04.2025

Flipped classroom – Q&A session

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“Flipped Classroom” Model

Lecture 3: Image Processing II (22.04.2021)

 [cv21-part03-image-processing2.pdf](#)
2MB Hochgeladen 19.04.2021 18:27

 [cv21-part03-image-processing2-6on1.pdf](#)
1.3MB Hochgeladen 19.04.2021 18:28

 Pre-recorded videos

"Flipped classroom" model: Please watch the following pre-recorded videos before the lecture on Thursday 22.04. The lecture slot will then contain a Q&A on the topics covered in the videos.

Pre-recorded videos

Part 2.3: Non-linear Filters



Computer Vision – Lecture 2

Image Processing I

2.3 Nonlinear Filters

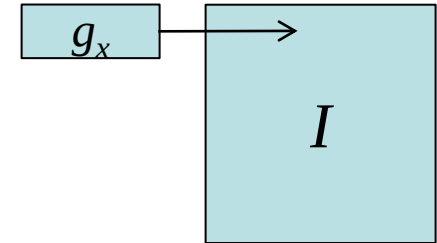
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- Pre-recorded videos made available on moodle
 - Part 2.3: Non-linear Filters
 - Part 3.1: Multi-Scale Representations
 - Part 3.2: Filters as Templates
 - Part 3.3: Image Derivatives
- *Quick poll: Who watched the videos?*

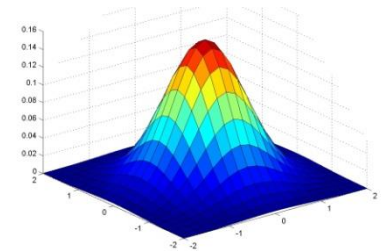


Course Outline

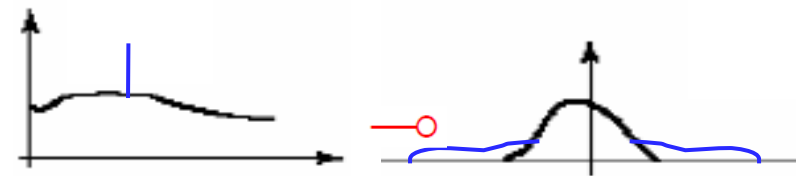
- Image Processing Basics
 - Image Formation
 - Linear Filters
 - Multi-scale Representations
 - Edge & Structure Extraction
- Segmentation & Grouping
- Object Recognition
- Local Features & Matching
- Deep Learning
- 3D Reconstruction



Linear Filters



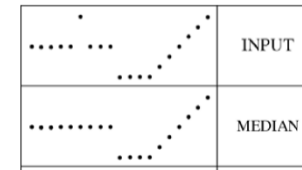
Gaussian Smoothing



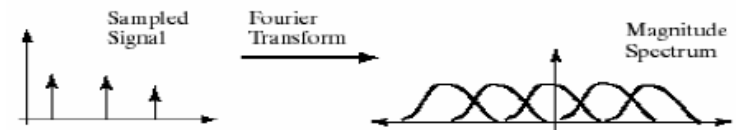
What does it mean to filter an image?

Course Outline

- Image Processing Basics
 - Image Formation
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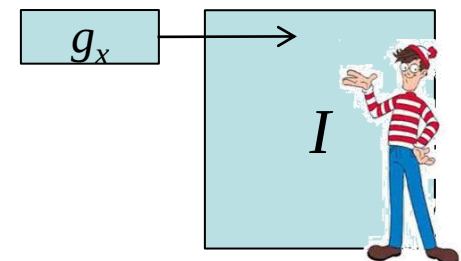
Non-Linear Filters



What does it mean to rescale an image?



Gaussian/Laplacian pyramid

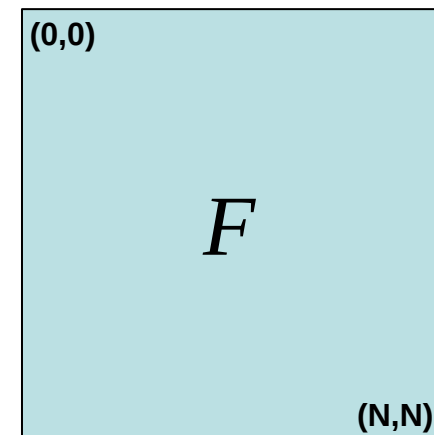
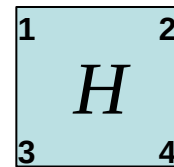


Filters as Templates

Recap: Correlation Filtering

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k H[u, v] F[i + u, j + v]$$

- This is called **cross-correlation**, denoted $G = H \otimes F$
- Filtering an image
 - Replace each pixel by a weighted combination of its neighbors.
 - The filter “kernel” or “mask” is the prescription for the weights in the linear combination.



Recap: Convolution

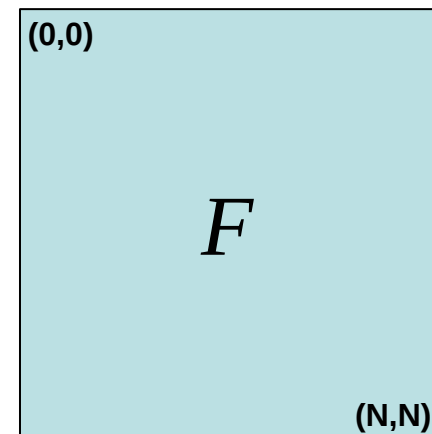
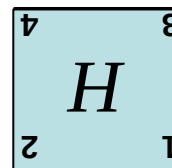
- Convolution:

- Flip the filter in both dimensions (bottom to top, right to left)
- Then apply cross-correlation

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k H[u, v] F[i - u, j - v]$$

$$G = H \star F$$

↑
*Notation for
convolution
operator*



Effect of Convolution

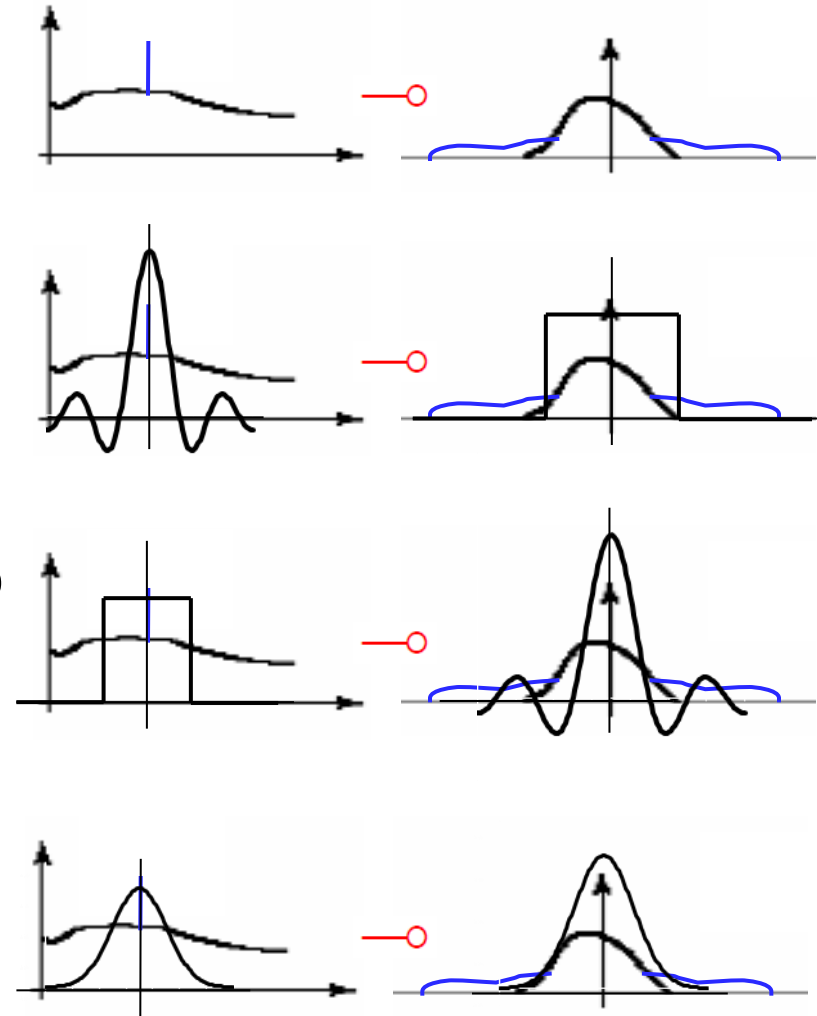
- Convoluting two functions in the image domain corresponds to taking the product of their transformed versions in the frequency domain.

$$f \star g \longrightarrow \mathcal{F} \cdot \mathcal{G}$$

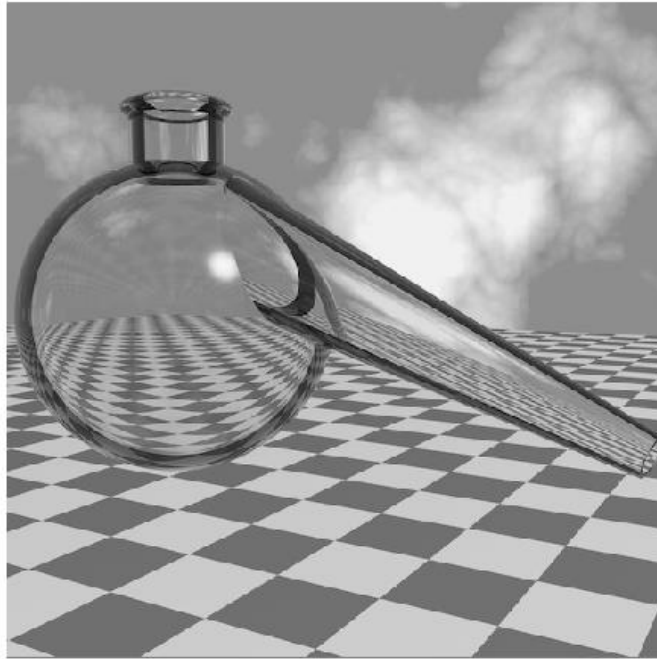
- This gives us a tool to manipulate image spectra.
 - A filter attenuates or enhances certain frequencies through this effect.

Recap: Effect of Filtering

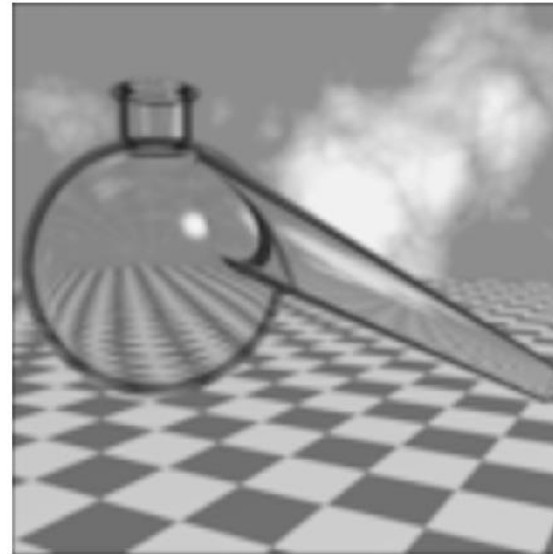
- Noise introduces high frequencies. To remove them, we want to apply a “low-pass” filter.
- The ideal filter shape in the frequency domain would be a box. But this transfers to a spatial sinc, which has infinite spatial support.
- A compact spatial box filter transfers to a frequency sinc, which creates artifacts.
- A Gaussian has compact support in both domains. This makes it a convenient choice for a low-pass filter.



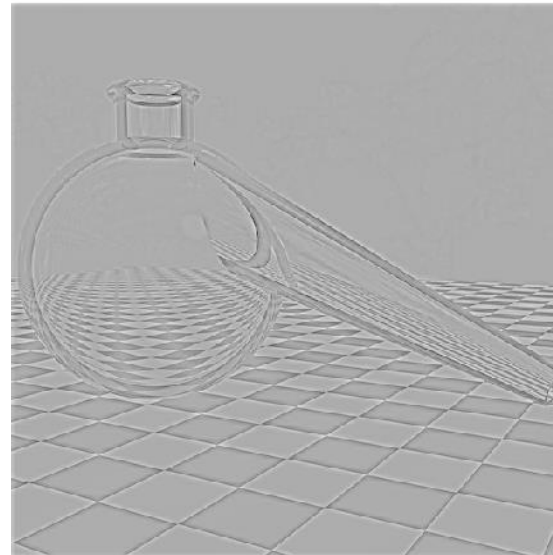
Low-Pass vs. High-Pass



Original image



Low-pass
filtered



High-pass
filtered

Topics of This Lecture

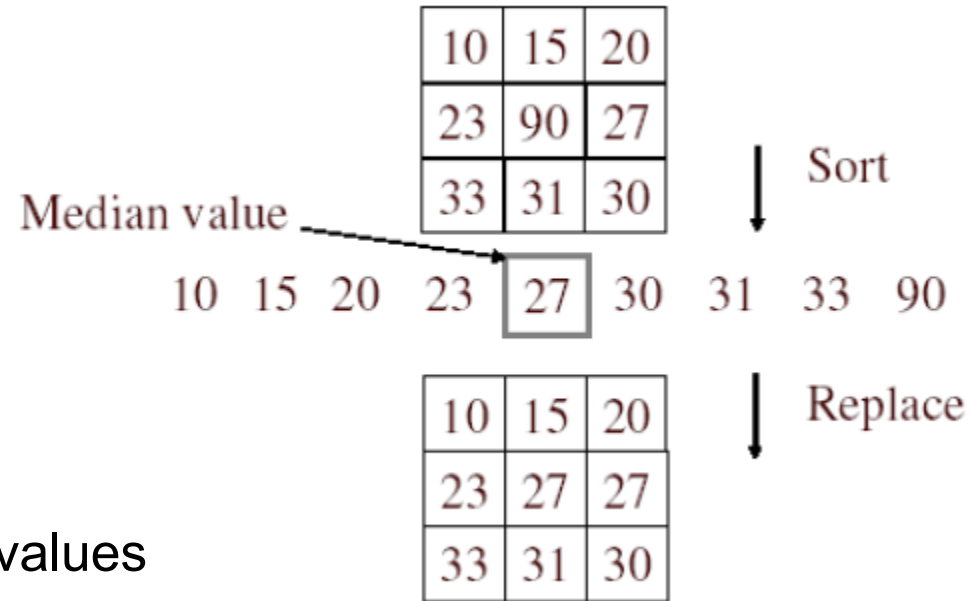
- Non-linear Filters
 - Median Filter
- Multi-Scale representations
 - How to properly rescale an image?
- Filters as templates
 - Correlation as template matching
- Image gradients
 - Derivatives of Gaussian



Recap: Median Filter (Non-Linear)

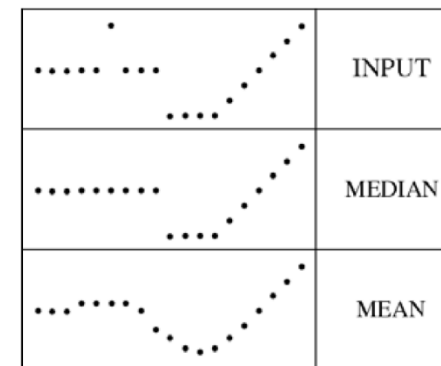
- Basic idea

- Replace each pixel by the median of its neighbors.



- Properties

- Doesn't introduce new pixel values
- Removes spikes: good for impulse, salt & pepper noise
- Edge preserving



Median vs. Gaussian Filtering

3x3

5x5

7x7

Gaussian



Median

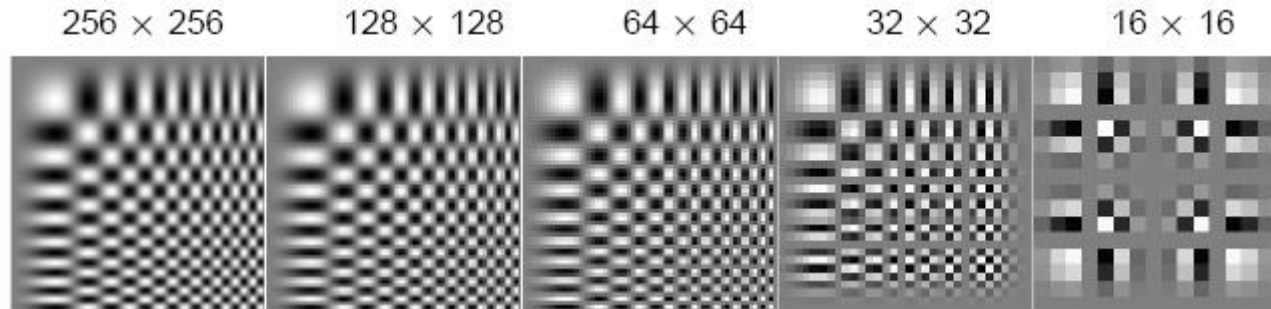


Topics of This Lecture

- Non-linear Filters
 - Median Filter
- Multi-Scale representations
 - How to properly rescale an image?
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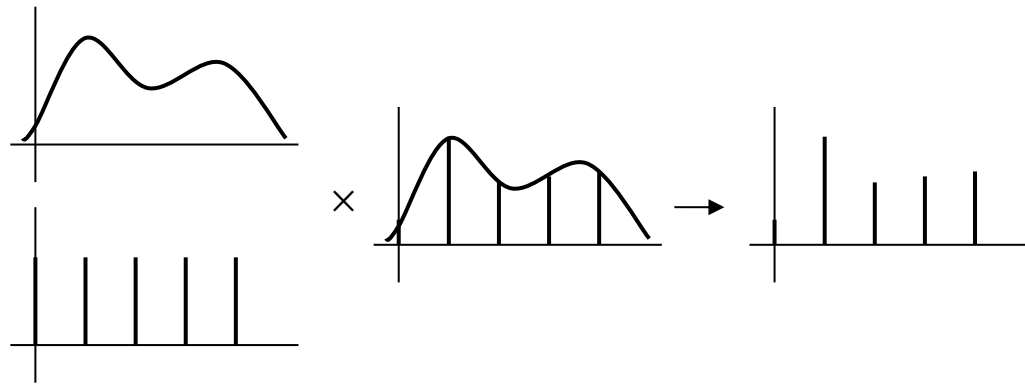
Recap: Multi-scale Representations



- Observation
 - When we downsample an image too much, artifacts will appear
- *Why is that?*
 - *And what is too much?*
 - For the answer, we again need to venture into the Fourier domain...

Fourier Interpretation: Discrete Sampling

- Sampling in the spatial domain is like multiplying with a spike function.

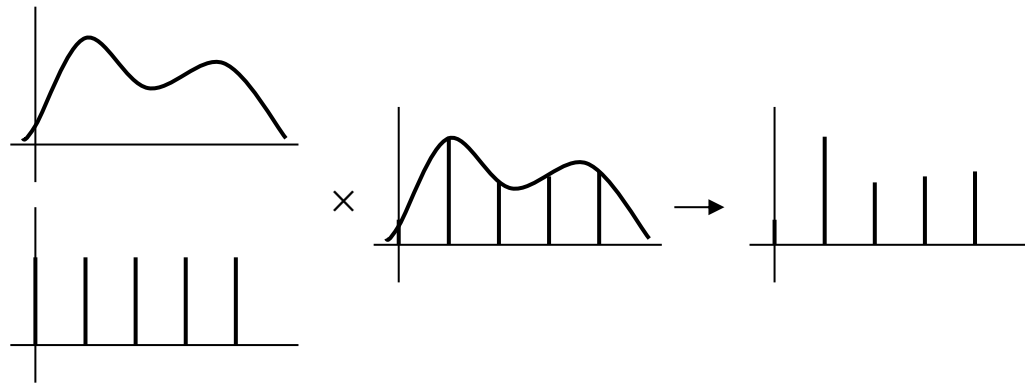


- Sampling in the frequency domain is like...

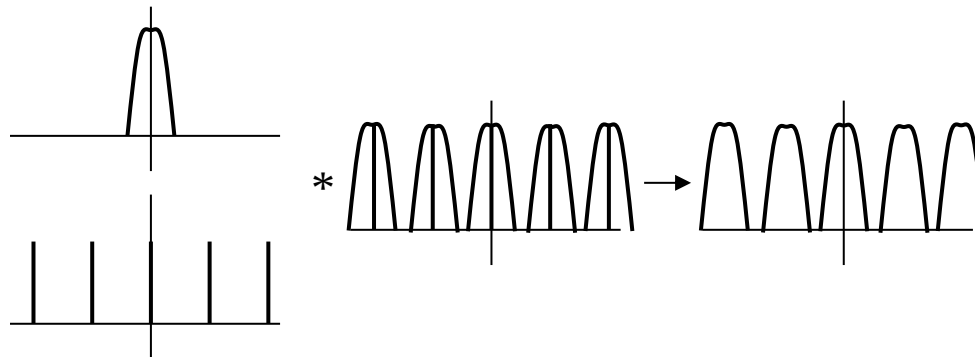
?

Fourier Interpretation: Discrete Sampling

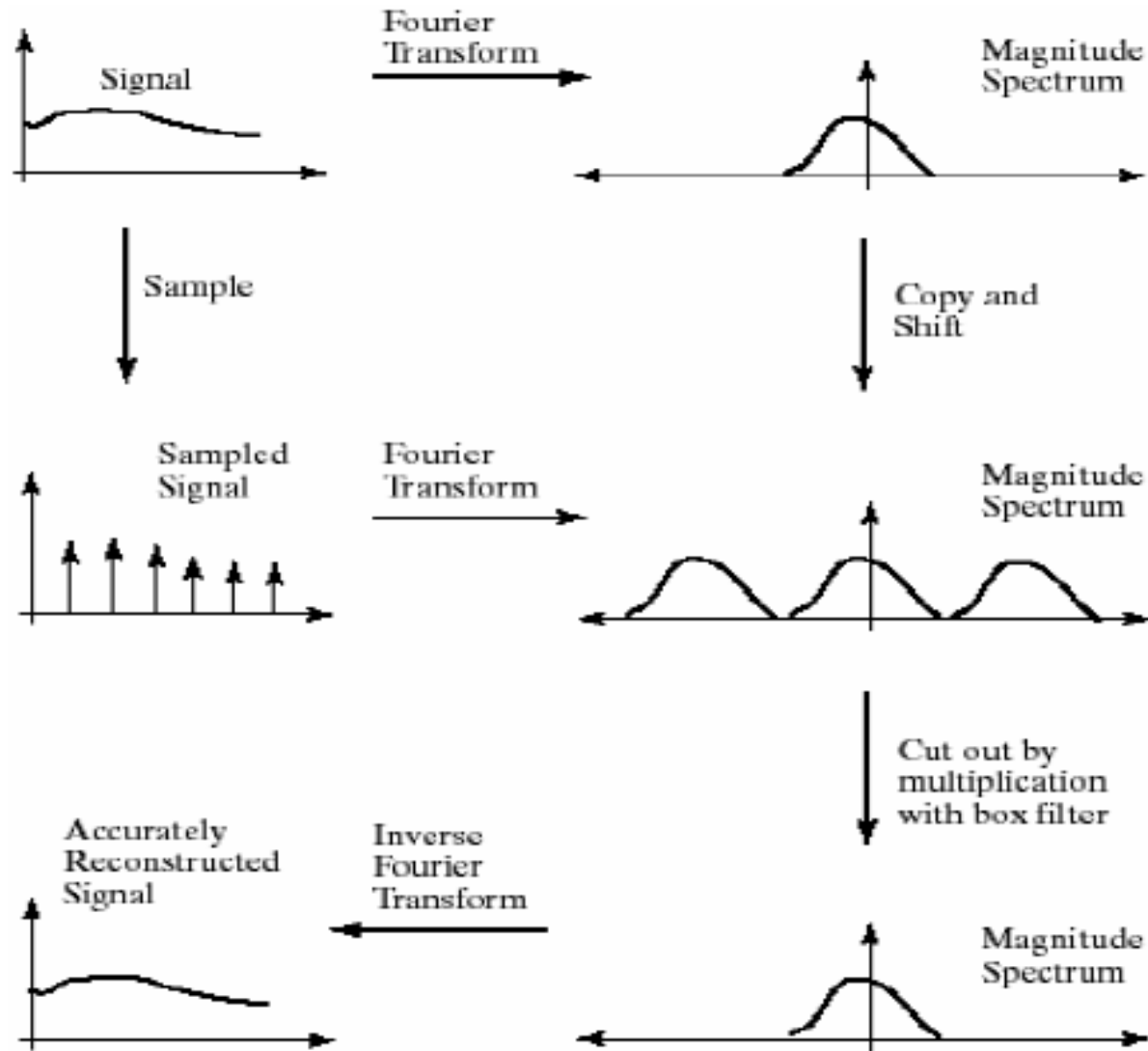
- Sampling in the spatial domain is like multiplying with a spike function.



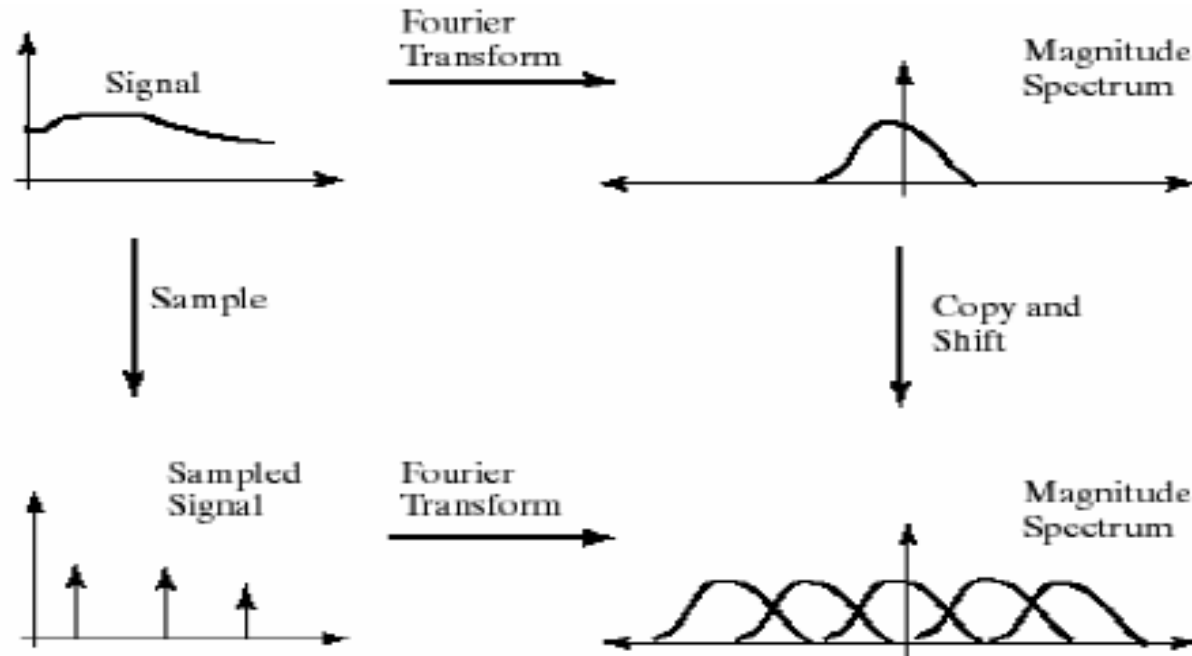
- Sampling in the frequency domain is like convolving with a spike function.



Recap: Sampling and Aliasing

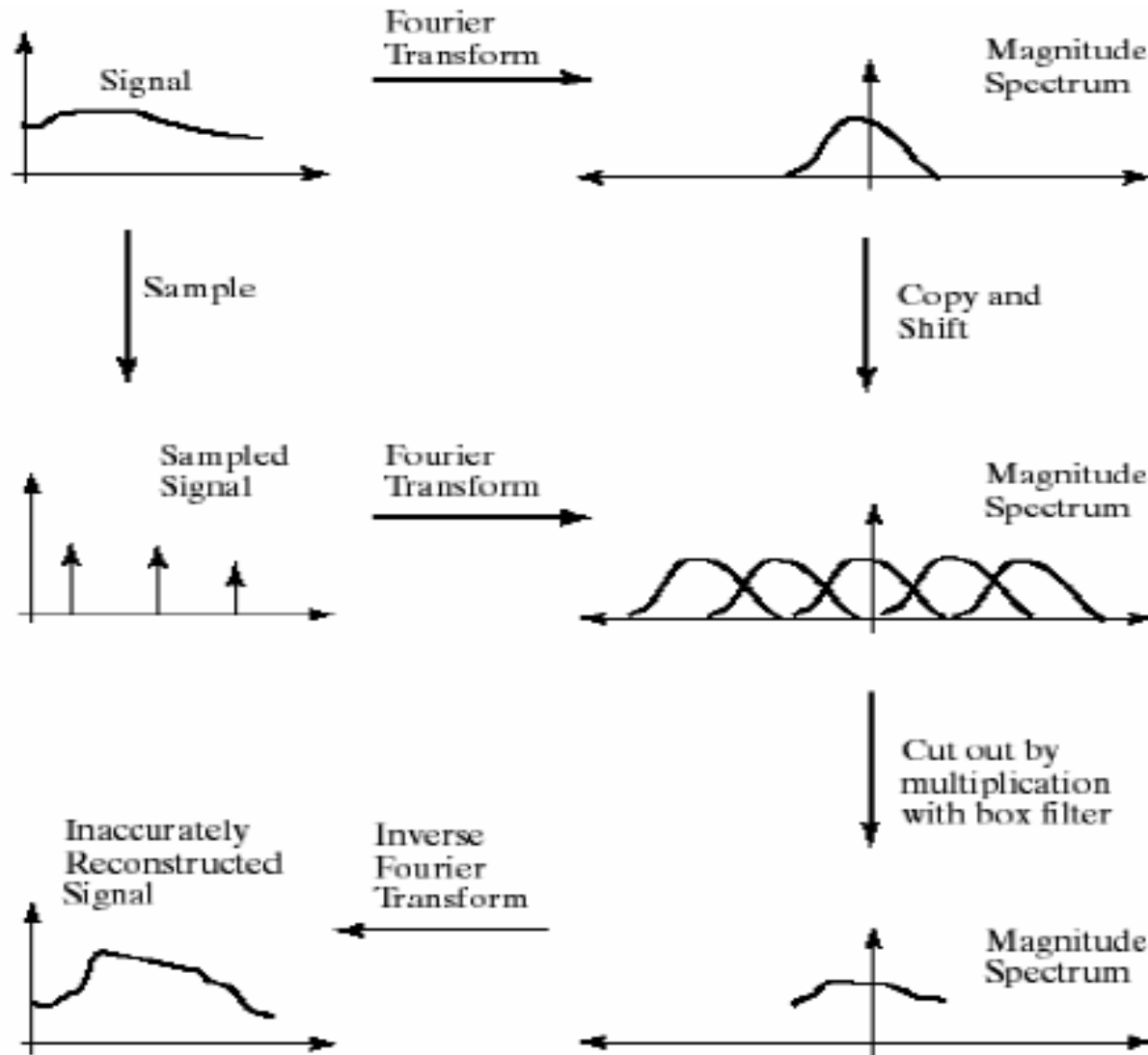


Recap: Sampling and Aliasing



- **Nyquist theorem:**
 - In order to recover a certain frequency f , we need to sample with at least $2f$.
 - This corresponds to the point at which the transformed frequency spectra start to overlap (the **Nyquist limit**)

Recap: Sampling and Aliasing

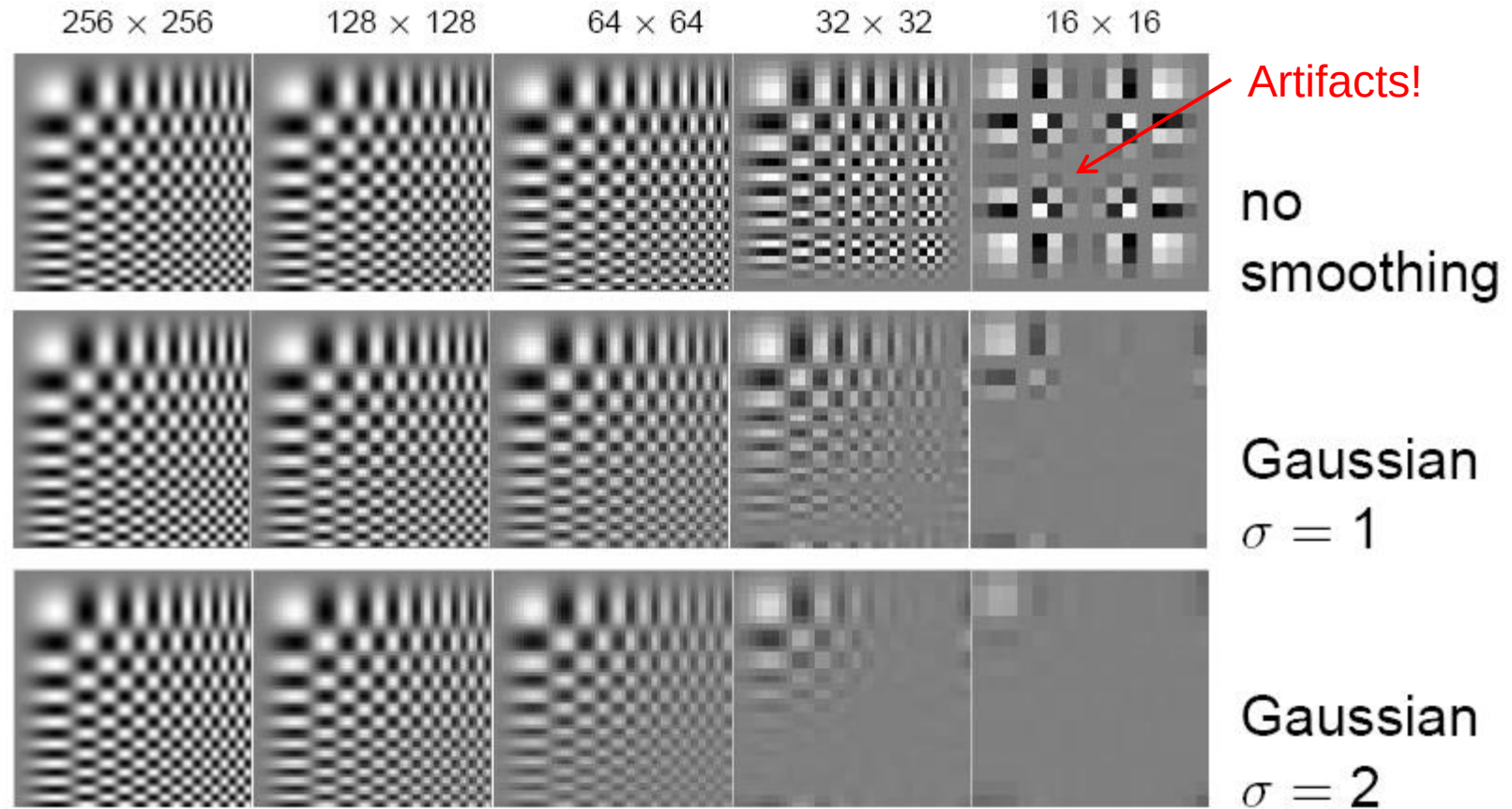


Aliasing in Graphics



Disintegrating textures

Recap: Resampling with Prior Smoothing



- Note: We cannot recover the high frequencies, but we can avoid artifacts by smoothing before resampling.

Recap: The Gaussian Pyramid

Low resolution

$$G_4 = (G_3 * \text{gaussian}) \downarrow 2$$

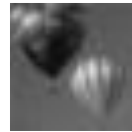
$$G_3 = (G_2 * \text{gaussian}) \downarrow 2$$

$$G_2 = (G_1 * \text{gaussian}) \downarrow 2$$

$$G_1 = (G_0 * \text{gaussian}) \downarrow 2$$

$$G_0 = \text{Image}$$

High resolution



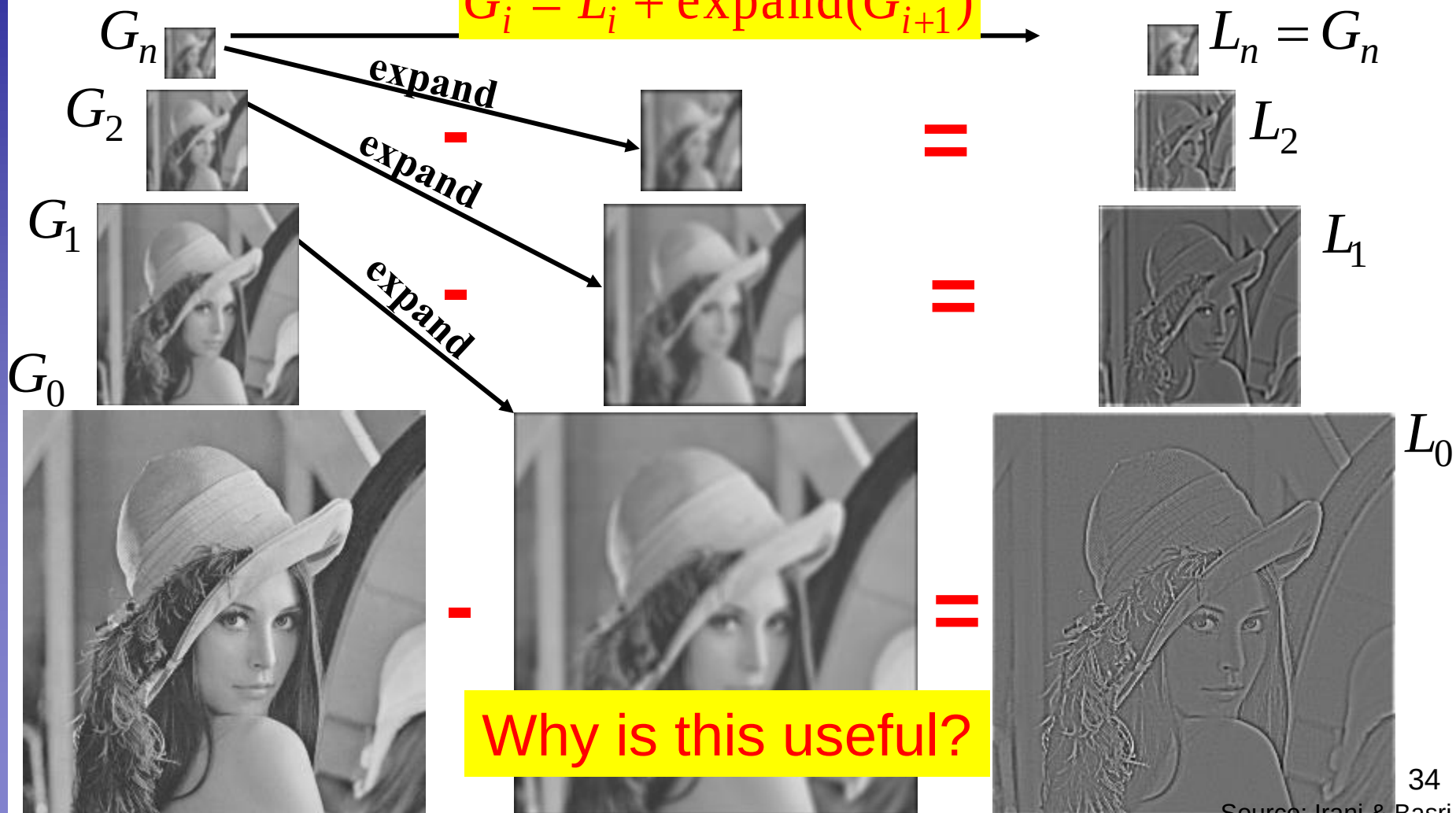
Recap: The Laplacian Pyramid

Gaussian Pyramid

$$L_i = G_i - \text{expand}(G_{i+1})$$

$$G_i = L_i + \text{expand}(G_{i+1})$$

Laplacian Pyramid



Why is this useful?

Topics of This Lecture

- Non-linear Filters
 - Median Filter
- Multi-Scale representations
 - How to properly rescale an image?
- Filters as templates
 - Correlation as template matching
- Image gradients
 - Derivatives of Gaussian



Where's Waldo?



Scene



Template

Where's Waldo?



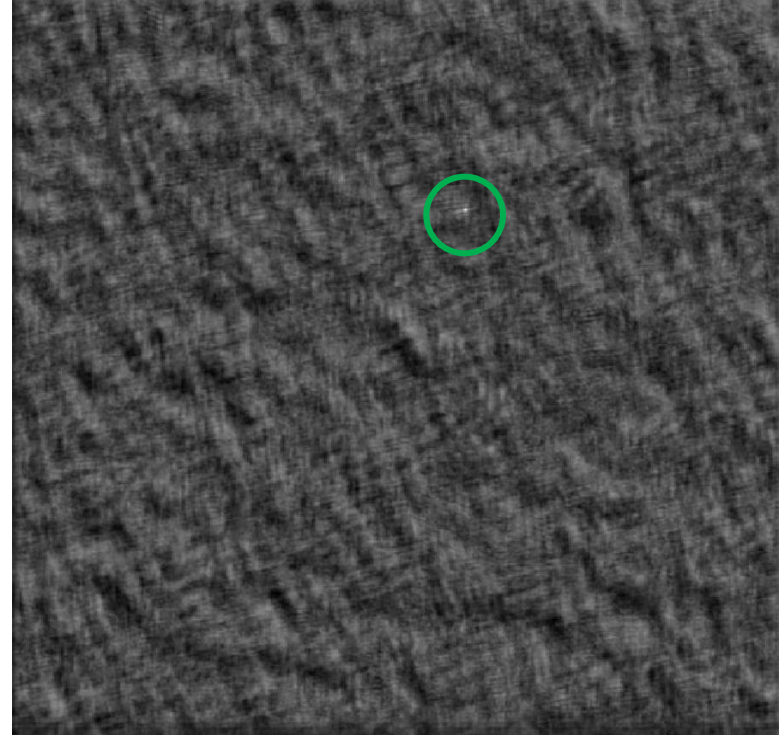
Template

Detected template

Where's Waldo?



Detected template



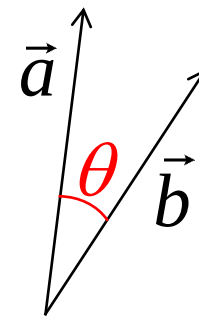
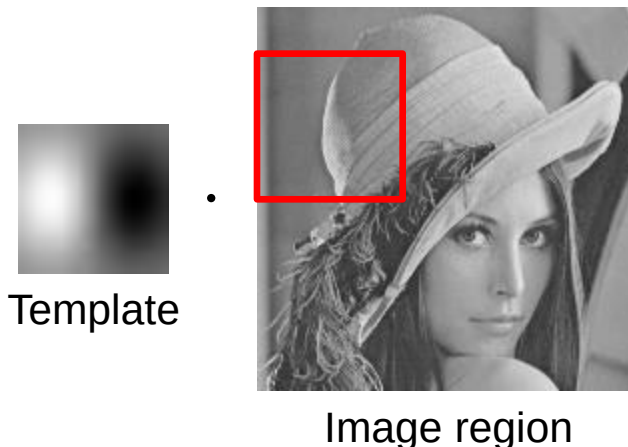
Correlation map

Recap: Correlation as Template Matching

- Think of filters as a dot product of the filter vector with the image region
 - Now measure the angle between the vectors

$$a \cdot b = |a| |b| \cos \theta \quad \cos \theta = \frac{a \cdot b}{|a| |b|}$$

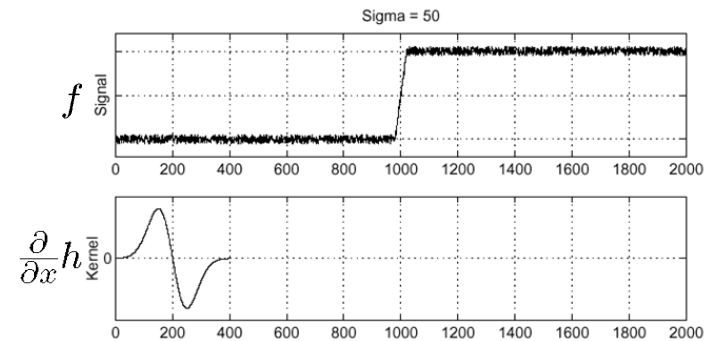
- Angle (similarity) between vectors can be measured by normalizing the length of each vector to 1 and taking the dot product.



Vector interpretation

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- Non-linear Filters
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Derivative Filters

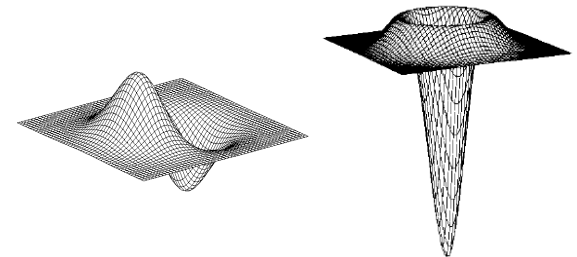
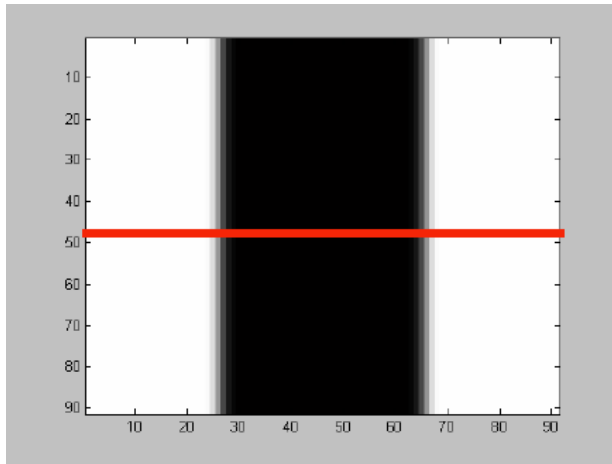
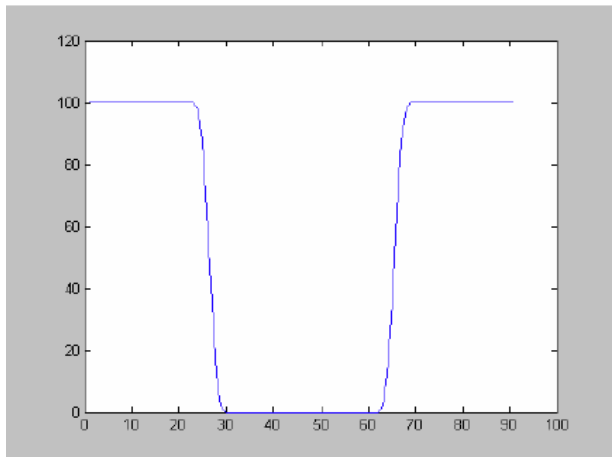
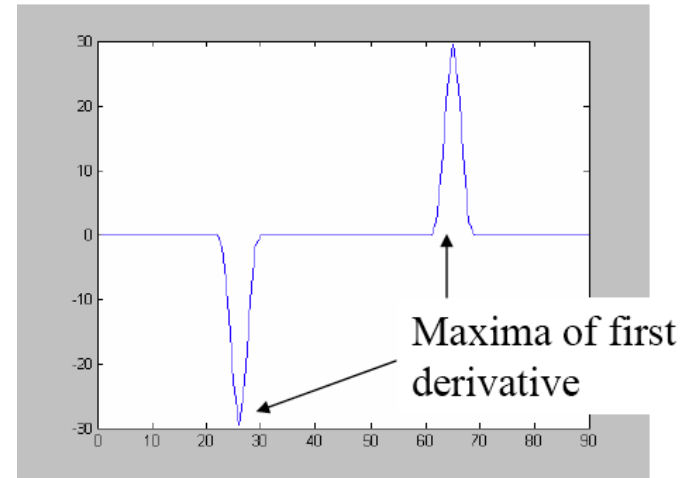


Image Derivative Operators

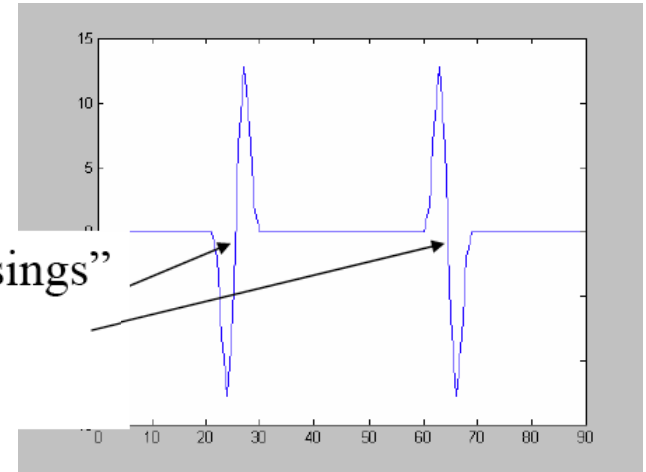
Recap: Derivatives and Edges...



1st derivative



2nd derivative



Recap: Finite Difference Filters

- For the 2D function $f(x,y)$, the partial derivative is
 - By definition:
$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \rightarrow 0} \frac{f(x+\varepsilon, y) - f(x, y)}{\varepsilon}$$
- Approximation for discrete data using **finite differences**

- Simplest case
$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x, y)}{1}$$
- More elaborate finite-difference approximations are available

- Implementation as a convolutional filter

- Filter mask:

1	-1
---	----

- In practice, use filters with an odd number of entries, e.g.,

Sobel

$$M_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

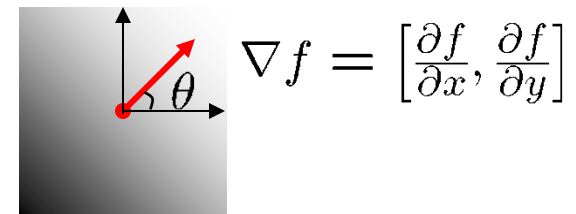
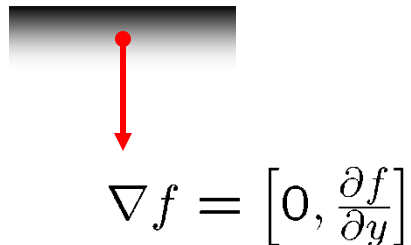
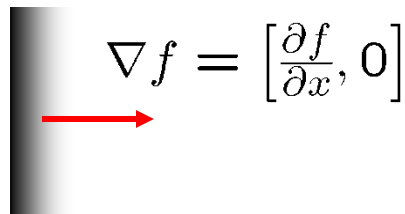
$$M_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

Recap: Image Gradient

- The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

- The gradient points in the direction of most rapid intensity change



- The gradient direction (orientation of edge normal) is given by:

$$\theta = \tan^{-1} \left(\frac{\partial f / \partial y}{\partial f / \partial x} \right)$$

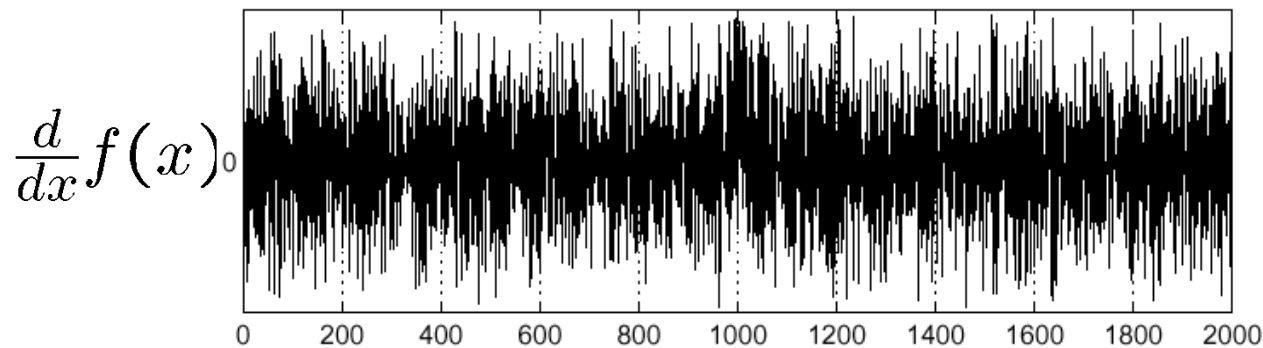
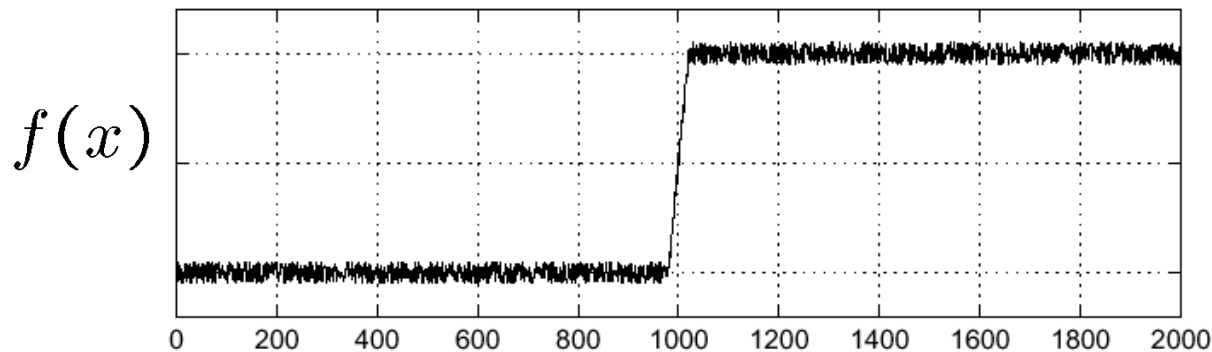
- The edge strength is given by the gradient magnitude

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2}$$



Recap: Effect of Noise

- Consider a single row or column of the image
 - Plotting intensity as a function of position gives a signal

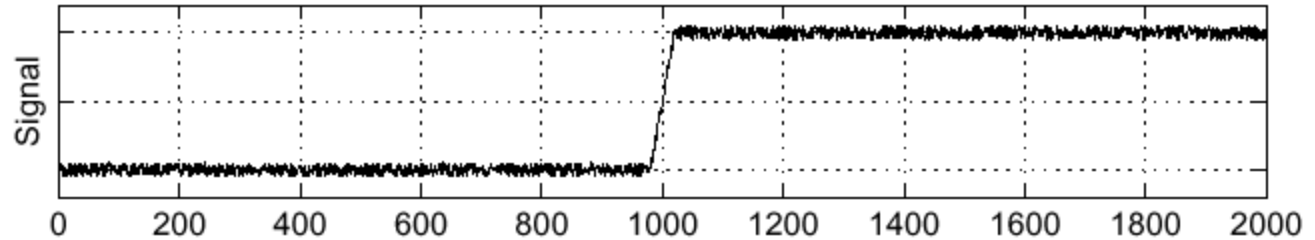


Where is the edge?

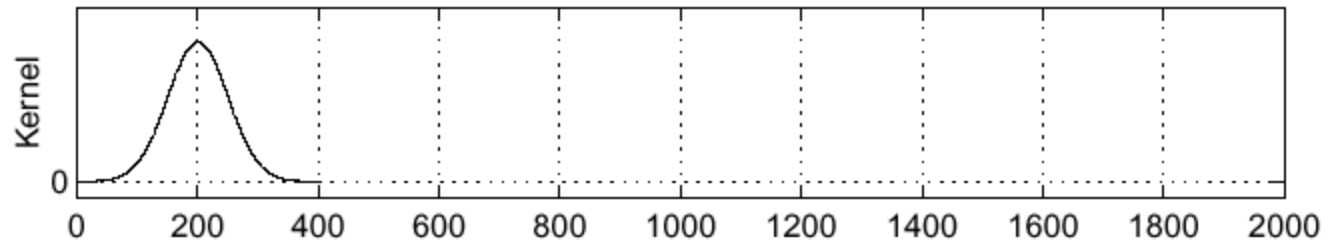
Recap: Solution: Smooth First

Sigma = 50

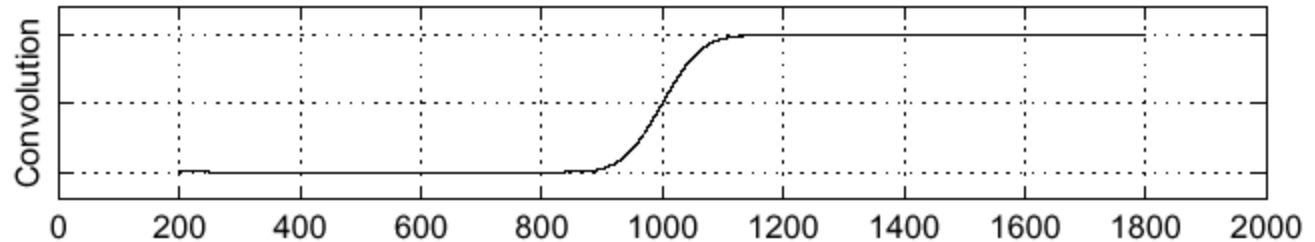
f



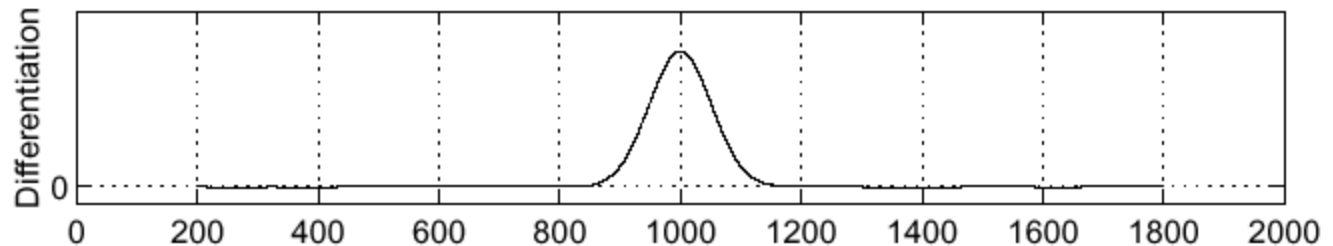
h



$h \star f$



$\frac{\partial}{\partial x}(h \star f)$



Where is the edge?

Look for peaks in $\frac{\partial}{\partial x}(h \star f)$

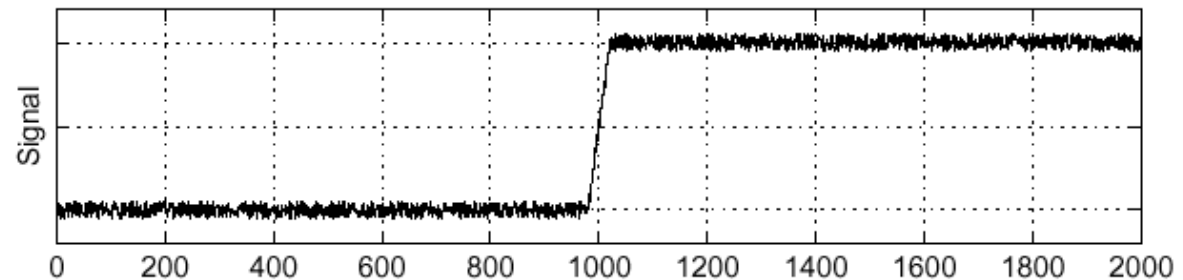
Recap: Derivative of Gaussian Filters

- Differentiation property of convolution.

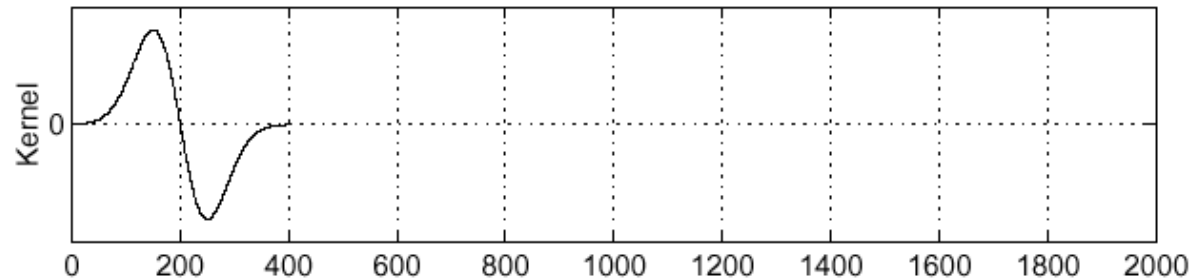
$$\frac{\partial}{\partial x}(h \star f) = \left(\frac{\partial}{\partial x}h\right) \star f$$

Sigma = 50

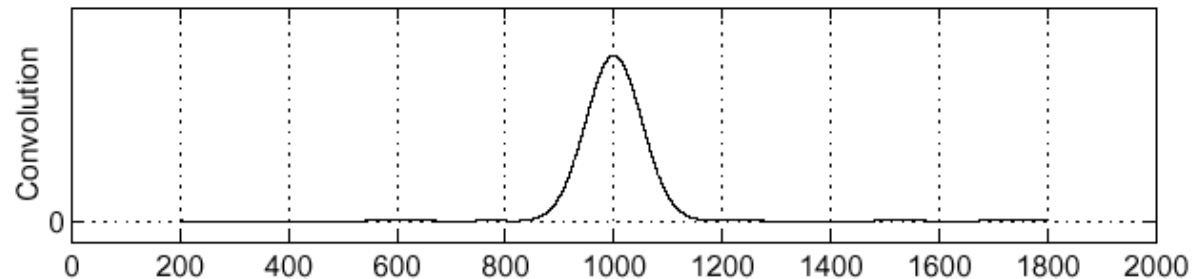
f



$\frac{\partial}{\partial x}h$



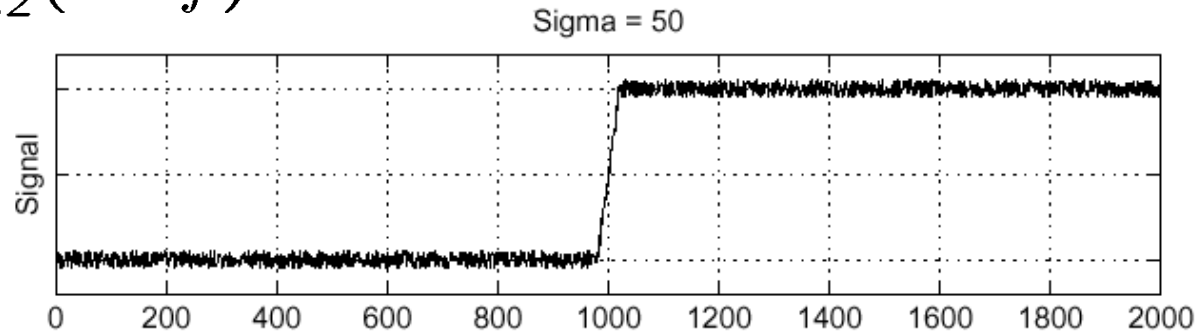
$\left(\frac{\partial}{\partial x}h\right) \star f$



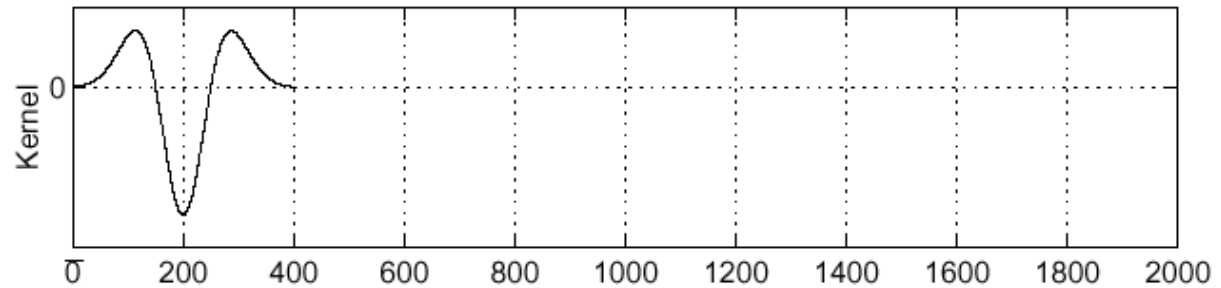
Recap: Laplacian of Gaussian (LoG)

- Consider $\frac{\partial^2}{\partial x^2}(h \star f)$

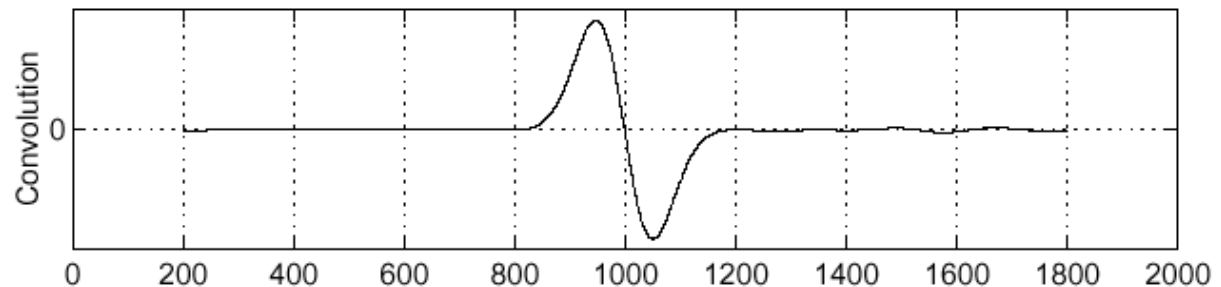
f



$\frac{\partial^2}{\partial x^2}h$



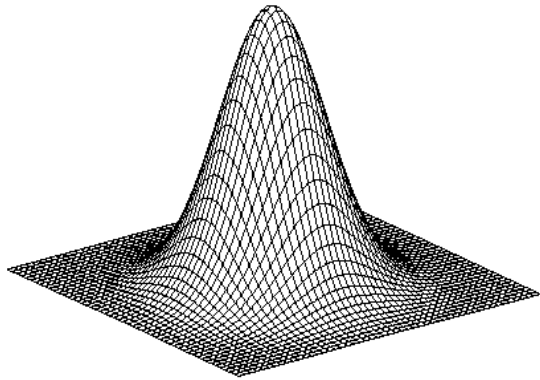
$(\frac{\partial^2}{\partial x^2}h) \star f$



Where is the edge?

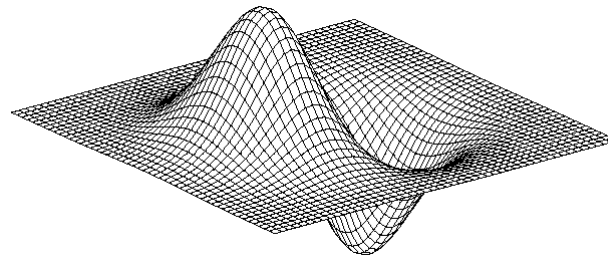
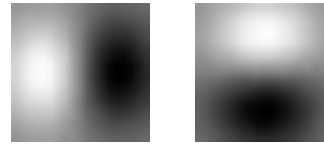
Zero-crossings of bottom graph

Recap: 2D Edge Detection Filters



Gaussian

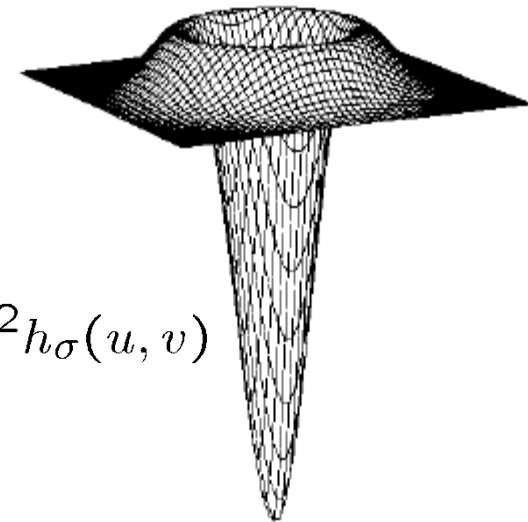
$$h_{\sigma}(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$$



Derivative of Gaussian

$$\frac{\partial}{\partial x} h_{\sigma}(u, v)$$

Laplacian of Gaussian

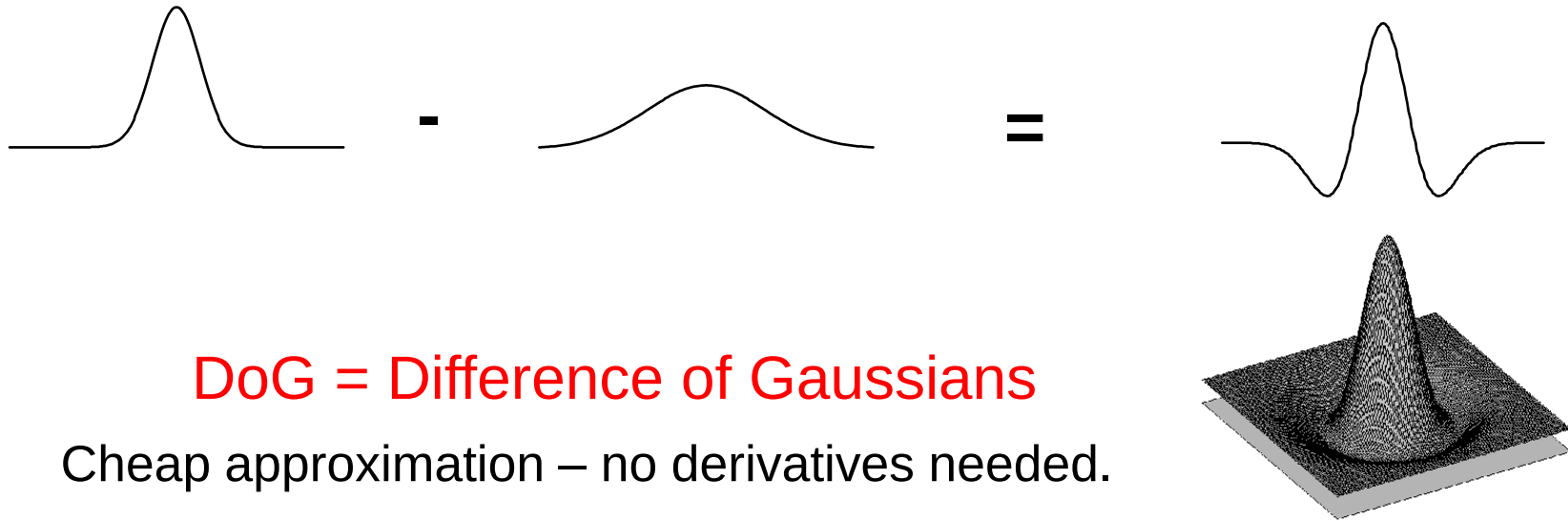


$$\nabla^2 h_{\sigma}(u, v)$$

- ∇^2 is the Laplacian operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

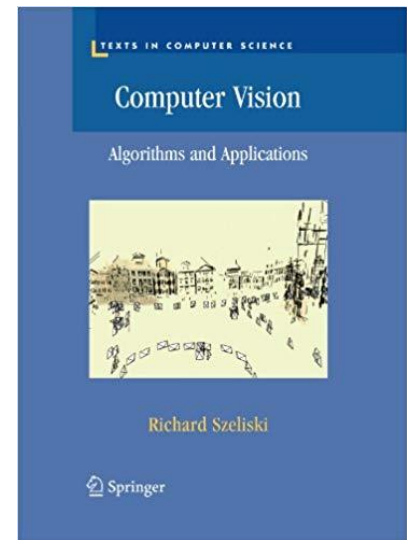
Recap: Laplacian $\approx$ Difference of Gaussian



References and Further Reading

- Background information on linear filters and image gradients can be found in Chapter 3 of the Szeliski book.

R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
([draft version available online](#))



Flipped Classroom Homework

- For the next lecture on Thursday next week, please watch the following pre-recorded video
 - Part 4.1: [Edge Detection](#)
- Remember: This week Thursday, we will have the first [Exercise](#).